

SILVER LAYER METADATA DICTIONARY

BI Portfolio Project

Version 1.0 | 29 Silver Layer Tables | SQL Server + Power BI
Retail Analytics — Kenya Market

29 Silver Tables	9 Business Domains	250+ Documented Columns	3 SCD-2 Entities
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About This Document

This document is the authoritative metadata dictionary for the Silver Layer of the BI project. The Silver Layer sits between the raw Bronze Layer (source data as-is) and the Gold Layer (analytics-ready tables). Every column in every silver table is described here — its data type, role (primary key, foreign key, or attribute), and business meaning.

This dictionary serves three purposes: (1) it provides the foundation for designing the Gold Layer in Document 2; (2) it acts as the technical reference for Power BI data modelling; and (3) it demonstrates data governance maturity to prospective employers.

Business Domain Reference

Domain	Description
Sales — POS	In-store Point-of-Sale transactions and line items
Sales — E-Commerce	Online orders and their line items
Customer	CRM master and customer profiles
Marketing	Campaigns and promotional programmes
Loyalty Programme	Points earn, redeem and balance tracking
Inventory	Stock movements and snapshot-level positions
Supply Chain	Purchase orders, goods receipts and supplier management
Human Resources	Employee records and time & attendance
Finance	General Ledger transactions and daily store P&L
Customer Experience	Feedback surveys and service interaction logs
Payments	Gift card issuance and redemption
Geospatial	County-level geographic and demographic reference
External / Macro	County-level macroeconomic indicators
Operations	Store master reference and lifecycle management
Market Intelligence	Competitor performance benchmarking data

Table Definitions

CAMPAIGNSMarketing

Stores marketing campaign definitions including budget allocation, channel, duration and spend tracking. Each row represents one marketing campaign run by the business.

Column	Role	Data Type	Description
campaign_id	PK	VARCHAR	Unique identifier for each campaign (e.g. CAMP-0001)
campaign_name	—	VARCHAR	Descriptive name of the campaign (e.g. 'Back to School 2024')
campaign_type	—	VARCHAR	High-level type: Seasonal, Product Launch, Loyalty, Clearance, etc.
channel	—	VARCHAR	Marketing channel used: Email, SMS, Social Media, In-Store, Radio, TV, etc.
start_date	—	DATE	Date the campaign officially begins
end_date	—	DATE	Date the campaign officially ends; NULL if open-ended
budget_kes	—	DECIMAL	Total budget allocated to the campaign in Kenyan Shillings
campaign_duration_days	—	INT	Calculated number of calendar days from start_date to end_date
actual_spend_kes	—	DECIMAL	Actual money spent on the campaign in KES (may differ from budget)
discount_rate	—	DECIMAL	Discount percentage offered to customers as part of this campaign (0–1 scale)

COMPETITOR_QUARTERLYMarket Intelligence

Quarterly snapshot of competitor financial and market-share metrics. Used to benchmark the business against external players. One row per competitor per quarter.

Column	Role	Data Type	Description
competitor_id	PK	VARCHAR	Unique identifier for each competitor entity
year	—	INT	Calendar year of the quarterly observation
qtr	—	VARCHAR	Quarter label: Q1, Q2, Q3, or Q4
revenue_kes	—	DECIMAL	Competitor's reported or estimated revenue in KES for that quarter
market_share_pct	—	DECIMAL	Competitor's estimated market share as a percentage (0–100)

CRMCustomer

Master customer record containing demographic, contact, behavioural and loyalty attributes. Each row represents one registered customer. This is the single source of truth for customer identity.

Column	Role	Data Type	Description
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customer_id	PK	VARCHAR	Unique customer identifier — joins to orders, feedback, loyalty, pos_transactions
first_name	—	VARCHAR	Customer's first (given) name
last_name	—	VARCHAR	Customer's last (family) name
gender	—	VARCHAR	Self-reported gender: Male, Female, Non-Binary, Prefer Not to Say
birth_date	—	DATE	Date of birth — used to calculate age and generation segments
phone	—	VARCHAR	Primary phone number (may include country code for Kenya: +254)
email	—	VARCHAR	Primary email address used for communications
county	—	VARCHAR	Kenyan county of residence — links to gis_counties for geospatial analysis
town	—	VARCHAR	Town or city within the county
customer_segment	—	VARCHAR	Business-defined segment: Premium, Regular, At-Risk, New, etc.
acquisition_channel	—	VARCHAR	How the customer was acquired: Online, In-Store, Referral, Social Media, etc.
registration_date	—	DATE	Date the customer created their account or was first registered
churn_date	—	DATE	Date the customer was marked as churned; NULL if still active
loyalty_tier	—	VARCHAR	Current loyalty programme tier: Bronze, Silver, Gold, Platinum
communication_preferences	—	VARCHAR	Preferred contact method: Email, SMS, WhatsApp, None, etc.
feedback_score	—	DECIMAL	Average feedback/NPS-style score from all submitted reviews (0–10 scale)

ECOMMERCE_ORDER_LINES

Sales — E-Commerce

Line-item detail for every e-commerce order. Each row represents one product line within an order, capturing pricing, discounts and quality flags applied during silver transformation.

Column	Role	Data Type	Description
order_line_key	PK	INT	Surrogate key uniquely identifying this order line record
order_id	FK	VARCHAR	Foreign key to ecommerce_orders — links line to its parent order
line_number	—	INT	Sequential line position within the order (1, 2, 3...)
product_id	FK	VARCHAR	Foreign key to products table
quantity	—	INT	Number of units of the product ordered on this line
unit_price_kes	—	DECIMAL	Original catalogue unit price in KES before any discounts

discount_rate	—	DECIMAL	Discount rate applied to this line (0–1 scale)
unit_price_after_discount_kes	—	DECIMAL	Effective unit price after applying the discount rate
discount_amount_kes	—	DECIMAL	Absolute KES value of the discount per unit
line_total_kes	—	DECIMAL	Total value of the line: unit_price_after_discount × quantity
calculated_line_total	—	DECIMAL	Cross-check recalculation of line_total for data quality validation
discount_tier	—	VARCHAR	Bucketed label for discount depth: No Discount, Low, Medium, High
order_prefix	—	VARCHAR	Prefix extracted from order_id (e.g. 'ORD') for pattern analysis
order_sequence_number	—	INT	Numeric sequence extracted from order_id for ordering/sorting
quality_flag	—	VARCHAR	Silver QA flag: PASS, WARN, or FAIL based on validation checks

ECOMMERCE_ORDERS

Sales — E-Commerce

Header-level record for each online order. Contains timing, customer linkage, delivery info, payment method and order status. One row per order.

Column	Role	Data Type	Description
order_id	PK	VARCHAR	Unique order identifier (e.g. ORD-00012)
order_date	—	DATE	Calendar date the order was placed
order_time	—	TIME	Time of day the order was placed
order_year	—	INT	Extracted year — pre-computed for faster filtering in Power BI
order_month	—	INT	Extracted month number (1–12)
order_quarter	—	VARCHAR	Quarter label: Q1–Q4
order_year_month	—	VARCHAR	YYYY-MM string for time-series grouping (e.g. '2024-03')
order_weekday	—	VARCHAR	Day of the week the order was placed: Monday–Sunday
order_hour	—	INT	Hour (0–23) extracted from order_time for hourly analysis
customer_id	FK	VARCHAR	Foreign key to crm table
delivery_address	—	VARCHAR	Full delivery address string
delivery_fee	—	DECIMAL	Delivery fee charged in KES; 0 if free delivery
payment_method	—	VARCHAR	Payment method used: M-Pesa, Card, Cash on Delivery, etc.

status	—	VARCHAR	Current order status: Pending, Processing, Shipped, Delivered, Cancelled, Returned
total_amount	—	DECIMAL	Total order value in KES including delivery fee

ECONOMIC

External / Macro

County-level macroeconomic indicators observed at monthly/quarterly intervals. Supports correlation analysis between economic conditions and retail performance.

Column	Role	Data Type	Description
economic_id	PK	INT	Surrogate key for the economic observation record
county	FK	VARCHAR	Kenyan county the observation applies to — links to gis_counties
date	—	DATE	Date of the observation (typically first day of the month)
qtr	—	VARCHAR	Quarter label (Q1–Q4) derived from the date
month	—	INT	Month number (1–12) derived from the date
gdp_growth_pct	—	DECIMAL	County-level estimated GDP growth rate as a percentage
inflation_pct	—	DECIMAL	Consumer Price Index-based inflation rate as a percentage
unemployment_pct	—	DECIMAL	Unemployment rate as a percentage of the labour force
consumer_confidence	—	DECIMAL	Index score reflecting consumer sentiment (higher = more optimistic)
retail_sales_index	—	DECIMAL	Index tracking retail sector sales volume relative to a base period
fuel_price_kes	—	DECIMAL	Average fuel price per litre in KES for the period
usd_kes_rate	—	DECIMAL	USD to KES exchange rate at the time of observation

FEEDBACK

Customer Experience

Customer feedback and review records with sentiment classification and date intelligence. Supports CSAT, NPS-style and sentiment trend analysis.

Column	Role	Data Type	Description
feedback_key	PK	INT	Surrogate key for the feedback record
feedback_id	—	VARCHAR	Business-facing feedback identifier
customer_id	FK	VARCHAR	Foreign key to crm table
feedback_date	—	DATE	Date the feedback was submitted
feedback_year	—	INT	Extracted year of submission

feedback_month	—	INT	Extracted month number of submission
feedback_quarter	—	VARCHAR	Quarter (Q1–Q4) the feedback falls in
feedback_year_month	—	VARCHAR	YYYY-MM string for time-series aggregation
rating	—	INT	Numeric rating provided by the customer (e.g. 1–5 stars)
rating_label	—	VARCHAR	Human-readable label: Poor, Fair, Good, Very Good, Excellent
sentiment	—	VARCHAR	NLP-derived sentiment classification: Positive, Neutral, Negative
category	—	VARCHAR	Feedback topic category: Product Quality, Delivery, Customer Service, Pricing, etc.
is_future_dated	—	BIT	Data quality flag: 1 if feedback_date is beyond today's date (anomaly)
has_no_comment	—	BIT	Flag: 1 if the customer submitted a rating without a written comment

GIFT_CARD_TRANSACTIONS

Payments

Individual debit/credit transactions against gift cards. Each row is one transaction event on a card.

Column	Role	Data Type	Description
transaction_id	PK	VARCHAR	Unique identifier for the gift card transaction
card_number	FK	VARCHAR	Foreign key to gift_cards table — identifies which card was used
date	—	DATE	Date the transaction occurred
amount	—	DECIMAL	Transaction value in KES (always positive)
type	—	VARCHAR	Transaction direction: Load (top-up) or Redemption (spend)

GIFT_CARDS

Payments

Master record for each issued gift card, including lifecycle dates and balance tracking.

Column	Role	Data Type	Description
card_number	PK	VARCHAR	Unique gift card number (also the FK target in gift_card_transactions)
issue_date	—	DATE	Date the card was issued to a customer
expiry_date	—	DATE	Date the card expires; NULL if it does not expire
initial_balance	—	DECIMAL	KES value loaded onto the card at time of issue
current_balance	—	DECIMAL	Remaining KES balance on the card after all transactions

GIS_COUNTIES

Geospatial

Reference table for Kenyan counties with demographic and geographic data. Enables map-based visualisations and county-level segmentation in Power BI.

Column	Role	Data Type	Description
county	PK	VARCHAR	County name — primary join key to crm, economic, stores tables
population	—	INT	Estimated residential population of the county
avg_income_kes	—	DECIMAL	Average annual household income in KES for the county
latitude	—	DECIMAL	Geographic latitude of the county centroid (for map plotting)
longitude	—	DECIMAL	Geographic longitude of the county centroid (for map plotting)

GL_TRANSACTIONS

Finance

General Ledger transaction records linking financial postings to accounts and stores. Supports P&L, balance sheet and store-level financial analysis.

Column	Role	Data Type	Description
gl_transaction_key	PK	INT	Surrogate key for the GL transaction record
transaction_id	—	VARCHAR	Business-facing GL transaction identifier
account_code	—	VARCHAR	Short code for the GL account (e.g. '4100')
account_number	—	VARCHAR	Full account number in the chart of accounts
account_name	—	VARCHAR	Descriptive name of the account (e.g. 'Sales Revenue')
account_category	—	VARCHAR	Account type: Revenue, Cost of Goods Sold, Operating Expense, Asset, Liability
store_id	FK	VARCHAR	Foreign key to stores table — which store this transaction belongs to
transaction_date	—	DATE	Date the financial transaction was posted
transaction_type	—	VARCHAR	Debit or Credit posting type
amount	—	DECIMAL	Absolute transaction amount in KES
signed_amount_debit_view	—	DECIMAL	Amount signed for a debit-normal view (debits positive, credits negative)
signed_amount_credit_view	—	DECIMAL	Amount signed for a credit-normal view (credits positive, debits negative)
transaction_year	—	INT	Extracted year of transaction date
transaction_month	—	INT	Extracted month number

transaction_quarter	—	VARCHAR	Quarter label (Q1–Q4)
transaction_year_month	—	VARCHAR	YYYY-MM string for period aggregation

GOODS_RECEIPTS

Supply Chain

Records of physical goods received from suppliers against purchase orders. Supports receiving performance, quality tracking and supplier reliability analysis.

Column	Role	Data Type	Description
receipt_key	PK	INT	Surrogate key for the receipt record
receipt_id	—	VARCHAR	Business-facing goods receipt identifier
po_number	FK	VARCHAR	Foreign key to purchase_orders — which PO this receipt fulfils
product_id	FK	VARCHAR	Foreign key to products — which product was received
receipt_date	—	DATE	Date goods were physically received at the store/warehouse
quantity_received	—	INT	Number of units received in this delivery
receiving_notes	—	VARCHAR	Free-text notes from the receiving team (damage, short delivery, etc.)
receipt_quality_category	—	VARCHAR	Categorised quality outcome: Good, Acceptable, Damaged, Short Delivery
has_quality_issue	—	BIT	Flag: 1 if the receipt has a quality issue, 0 otherwise
receipt_year	—	INT	Extracted year of receipt_date
receipt_month	—	INT	Extracted month number
receipt_quarter	—	VARCHAR	Quarter label (Q1–Q4)
receipt_year_month	—	VARCHAR	YYYY-MM string for period grouping

HR

Human Resources

Comprehensive employee master table with demographics, job details, salary information, tenure metrics and derived HR analytics attributes. Supports workforce dashboards and people analytics.

Column	Role	Data Type	Description
employee_id	PK	VARCHAR	Unique employee identifier
first_name	—	VARCHAR	Employee's first name
last_name	—	VARCHAR	Employee's last (family) name
gender	—	VARCHAR	Self-reported gender

job_title	—	VARCHAR	Formal job title (e.g. Store Manager, Cashier, Warehouse Associate)
store_id	FK	VARCHAR	Foreign key to stores — the store this employee is assigned to
shift_pattern	—	VARCHAR	Work shift pattern: Morning, Afternoon, Evening, Night, Rotating
valid_from	—	DATE	SCD effective start date — when this version of the record became active
valid_to	—	DATE	SCD effective end date — NULL if this is the current active record
birth_date	—	DATE	Employee date of birth
hire_date	—	DATE	Date the employee was officially hired
salary	—	DECIMAL	Current monthly gross salary in KES
education_level	—	VARCHAR	Highest education attained: Primary, Secondary, Diploma, Degree, Postgraduate
age	—	INT	Calculated age in years as of the silver layer load date
generation	—	VARCHAR	Generational cohort: Gen Z, Millennial, Gen X, Boomer
full_name	—	VARCHAR	Concatenated first_name + last_name
display_name	—	VARCHAR	Formatted display name (e.g. 'J. Kamau') for reports
generated_email	—	VARCHAR	System-generated work email derived from name and employee_id
tenure_days	—	INT	Number of days from hire_date to silver load date (or valid_to)
tenure_months	—	INT	tenure_days expressed in whole months
tenure_years	—	DECIMAL	tenure_days expressed in decimal years
tenure_band	—	VARCHAR	Bucketed tenure label: <1 Year, 1–3 Years, 3–5 Years, 5+ Years
retention_risk	—	VARCHAR	Derived retention risk score: Low, Medium, High — based on tenure & other signals
employment_status	—	VARCHAR	Current status: Active, On Leave, Terminated, Resigned
contract_length_months	—	INT	Length of employment contract in months (NULL for permanent roles)
contract_type	—	VARCHAR	Type of contract: Permanent, Fixed-Term, Part-Time, Casual
salary_band	—	VARCHAR	Salary bracket label: Band 1 (lowest) through Band 5 (highest)
salary_equity_flag	—	VARCHAR	Flag comparing employee salary to peers in same role/band: Fair, Below, Above
department_name	—	VARCHAR	Department the employee belongs to: Operations, Finance, HR, IT, etc.
department_category	—	VARCHAR	Broader department grouping: Front-of-House, Back-of-House, Corporate

job_level	—	VARCHAR	Hierarchical job level: Junior, Mid, Senior, Lead, Manager, Director
is_manager	—	BIT	Flag: 1 if the employee has a management role
hire_month	—	INT	Month number of hire_date — for cohort and seasonal hiring analysis
hire_quarter	—	VARCHAR	Quarter of hire (Q1–Q4)
hire_season	—	VARCHAR	Season of hire: Summer, Autumn, Winter, Spring
hire_year	—	INT	Year of hire — for cohort analysis
years_since_hire	—	DECIMAL	Precise number of years from hire_date to load date
hire_cohort	—	VARCHAR	Grouped hire year cohort label (e.g. '2020–2021')
gender_category	—	VARCHAR	Simplified gender grouping for diversity reporting
age_band	—	VARCHAR	Age bracket: Under 25, 25–34, 35–44, 45–54, 55+

INVENTORY_MOVEMENTS

Inventory

Granular record of every stock movement event (receipts, sales, adjustments, transfers) across stores and products. Enables inventory flow analysis, demand velocity and shrinkage tracking.

Column	Role	Data Type	Description
inventory_movement_key	PK	INT	Surrogate key for the movement record
movement_id	—	VARCHAR	Business-facing movement identifier
movement_date	—	DATE	Date the inventory movement occurred
store_id	FK	VARCHAR	Foreign key to stores — which store the movement happened at
product_id	FK	VARCHAR	Foreign key to products — which product moved
movement_type	—	VARCHAR	Type of movement: Receipt, Sale, Adjustment, Transfer In, Transfer Out, Return
quantity	—	INT	Signed quantity: positive for inflows, negative for outflows
quantity_absolute	—	INT	Absolute (unsigned) quantity for volume analysis
unit_cost_kes	—	DECIMAL	Cost per unit in KES at time of movement
movement_value_kes	—	DECIMAL	Total value of the movement (quantity_absolute × unit_cost_kes)
running_quantity	—	INT	Running total stock quantity for the product across all stores
cumulative_sum_by_product	—	INT	Cumulative quantity moved for the product across all time
cumulative_sum_by_product_store	—	INT	Cumulative quantity moved for product at a specific store

running_total_raw_quantity	—	INT	Raw running total before any sign correction
moving_sum_3_transactions	—	INT	Moving sum of quantity over the last 3 transactions (smoothing)
inventory_status	—	VARCHAR	Derived status after movement: Adequate, Low, Overstock, Out of Stock
demand_velocity	—	VARCHAR	Derived label for how fast this product moves: Slow, Medium, Fast
movement_year	—	INT	Extracted year of movement_date
movement_month	—	INT	Extracted month number
movement_quarter	—	VARCHAR	Quarter label (Q1–Q4)
movement_year_month	—	VARCHAR	YYYY-MM string for period aggregation
movement_month_name	—	VARCHAR	Full month name (e.g. 'March') for display
sign_validation_flag	—	VARCHAR	Quality check: PASS if quantity sign matches movement_type, FAIL otherwise
quality_flag	—	VARCHAR	Overall silver QA flag: PASS, WARN, or FAIL

INVENTORY_SNAPSHOTS

Inventory

Point-in-time snapshot of on-hand stock levels per store per product. Used for stock status reporting, reorder analysis and inventory health dashboards.

Column	Role	Data Type	Description
snapshot_key	PK	INT	Surrogate key for the snapshot record
store_id	FK	VARCHAR	Foreign key to stores table
product_id	FK	VARCHAR	Foreign key to products table
snapshot_date	—	DATE	Date the snapshot was taken
on_hand_quantity	—	INT	Physical units in stock at time of snapshot
reorder_point	—	INT	Minimum quantity threshold that triggers a reorder
safety_stock	—	INT	Buffer stock to be maintained above the reorder point
stock_vs_reorder	—	INT	Difference: on_hand_quantity minus reorder_point (negative = needs reorder)
stock_status	—	VARCHAR	Derived status: Adequate, Low, Critical, Overstock
suggested_reorder_quantity	—	INT	Calculated units to order to restore stock to optimal level
snapshot_year	—	INT	Extracted year of snapshot_date
snapshot_month	—	INT	Extracted month number
snapshot_quarter	—	VARCHAR	Quarter label (Q1–Q4)

snapshot_year_month	—	VARCHAR	YYYY-MM for period grouping
snapshot_weekday	—	VARCHAR	Day of week the snapshot was taken
quality_flag	—	VARCHAR	Silver QA flag: PASS, WARN, or FAIL

LOYALTY_TRANSACTIONS

Loyalty Programme

Records every points earn and redemption event in the loyalty programme. Enables points balance tracking, customer engagement analysis and programme ROI reporting.

Column	Role	Data Type	Description
loyalty_transaction_key	PK	INT	Surrogate key for this loyalty transaction record
transaction_id	—	VARCHAR	Business-facing loyalty transaction identifier
customer_id	FK	VARCHAR	Foreign key to crm table
order_id	FK	VARCHAR	Foreign key to ecommerce_orders or pos_transactions — the purchase that triggered this event
transaction_date	—	DATE	Date the loyalty event occurred
transaction_type	—	VARCHAR	Event type: Earn, Redeem, Expire, Adjustment, Bonus
points_change	—	INT	Net points change on this transaction (positive = earn, negative = redeem/expire)
points_earned_raw	—	INT	Raw points earned before any adjustments or multipliers
points_redeemed	—	INT	Points redeemed on this transaction (0 if not a redemption event)
points_balance	—	INT	Running points balance after this transaction
order_source_type	—	VARCHAR	Source channel of the originating order: E-Commerce, POS, Other
points_activity_tier	—	VARCHAR	Tier label based on points change size: Low, Medium, High Activity
customer_point_status	—	VARCHAR	Customer loyalty standing at time of transaction: Active Earner, Redeemer, Lapsing, Inactive
transaction_year	—	INT	Extracted year of transaction_date
transaction_month	—	INT	Extracted month number
transaction_quarter	—	VARCHAR	Quarter label (Q1–Q4)
transaction_year_month	—	VARCHAR	YYYY-MM string for period aggregation
quality_flag	—	VARCHAR	Silver QA flag: PASS, WARN, or FAIL
etl_load_date	—	DATETIME	Timestamp when this record was loaded into the silver layer
etl_source	—	VARCHAR	Source system identifier (e.g. 'loyalty_system_v2')

POS_LINE_ITEMS

Sales — POS

Line-item detail for every in-store Point-of-Sale transaction. Each row is one product sold within a transaction, including pricing and discount details.

Column	Role	Data Type	Description
pos_line_key	PK	INT	Surrogate key for this POS line item
transaction_id	FK	VARCHAR	Foreign key to pos_transactions
line_number	—	INT	Sequential line position within the transaction
product_id	FK	VARCHAR	Foreign key to products table
quantity	—	INT	Units sold on this line
unit_price_kes	—	DECIMAL	Original shelf price per unit in KES
discount_rate	—	DECIMAL	Discount applied to this line (0–1 scale)
effective_unit_price_kes	—	DECIMAL	Unit price after discount: $\text{unit_price_kes} \times (1 - \text{discount_rate})$
discount_amount_kes	—	DECIMAL	Absolute discount value per unit in KES
line_total_kes	—	DECIMAL	Total line value: $\text{effective_unit_price} \times \text{quantity}$
calculated_line_total	—	DECIMAL	QA recalculation of line_total for cross-check validation
discount_tier	—	VARCHAR	Discount depth label: No Discount, Low, Medium, High
line_value_tier	—	VARCHAR	Line total value bracket: Low Value, Medium, High Value
transaction_source	—	VARCHAR	Origin of the transaction: POS Terminal, Self-Checkout, Mobile POS

POS_TRANSACTIONS

Sales — POS

Header record for every in-store Point-of-Sale transaction. Links the sale to a store, cashier, customer (if known) and payment method.

Column	Role	Data Type	Description
transaction_id	PK	VARCHAR	Unique POS transaction identifier
date	—	DATE	Date the transaction occurred
time	—	TIME	Time the transaction was completed
store_id	FK	VARCHAR	Foreign key to stores table
customer_id	FK	VARCHAR	Foreign key to crm — NULL if the customer was not identified (walk-in)

cashier_id	FK	VARCHAR	Foreign key to hr — the employee who processed the transaction
payment_method	—	VARCHAR	Payment method: M-Pesa, Cash, Card (Debit/Credit), Gift Card
total_amount	—	DECIMAL	Total transaction value in KES

PRODUCTS

Product Catalogue

Product master reference with pricing, categorisation and SCD lifecycle management. One row per active product version; discontinued products remain for historical join accuracy.

Column	Role	Data Type	Description
product_id	PK	VARCHAR	Unique product identifier — joins to all sales, inventory and supply chain tables
supplier_id	FK	VARCHAR	Foreign key to suppliers — who supplies this product
category	—	VARCHAR	Top-level product category (e.g. Beverages, Dairy, Electronics)
subcategory	—	VARCHAR	More granular classification within the category
product_name	—	VARCHAR	Full product display name
brand	—	VARCHAR	Brand name of the product
introduction_date	—	DATE	Date the product was first added to the catalogue
valid_from	—	DATE	SCD effective start date for this price/cost version
valid_to	—	DATE	SCD effective end date; NULL if current active version
discontinued_date	—	DATE	Date the product was discontinued; NULL if still active
is_active	—	BIT	Flag: 1 if product is currently available for sale
unit_cost_kes	—	DECIMAL	Cost to the business per unit in KES
retail_price_kes	—	DECIMAL	Standard retail selling price per unit in KES
margin_percentage	—	DECIMAL	Gross margin as a percentage: $(\text{retail_price} - \text{unit_cost}) / \text{retail_price} \times 100$

PROMOTION_PRODUCTS

Marketing

Bridge/junction table linking promotions to specific products. Resolves the many-to-many relationship between promotions and the products they apply to.

Column	Role	Data Type	Description
promotion_product_key	PK	INT	Surrogate key for this promotion-product association
promotion_id	FK	VARCHAR	Foreign key to promotions table

product_id	FK	VARCHAR	Foreign key to products table
record_type	—	VARCHAR	Classification of how the product is included: Discount, Bundle, Gift With Purchase, etc.

PROMOTIONS

Marketing

Master record for every promotional event. Tracks lifecycle, discount mechanism and current status. Links to [promotion_products](#) for product-level details.

Column	Role	Data Type	Description
promotion_key	PK	INT	Surrogate key for the promotion record
promotion_id	—	VARCHAR	Business-facing promotion identifier
promotion_name	—	VARCHAR	Full descriptive name of the promotion
promotion_short_name	—	VARCHAR	Abbreviated name for use in report labels
start_date	—	DATE	Date the promotion begins
end_date	—	DATE	Date the promotion ends
campaign_duration_days	—	INT	Number of calendar days the promotion runs
promotion_status	—	VARCHAR	Current status: Upcoming, Active, Expired
campaign_length_tier	—	VARCHAR	Duration bucket: Short (<7 days), Medium (7–30 days), Long (30+ days)
discount_type	—	VARCHAR	Mechanism: Percentage Off, Fixed Amount Off, BOGO, Bundle, Free Item
discount_value	—	DECIMAL	Numeric value of the discount (percentage or fixed amount depending on type)
discount_amount_or_percent	—	VARCHAR	Clarifies whether discount_value is an Amount (KES) or Percent (%)
discount_description	—	VARCHAR	Human-readable description of the discount (e.g. '20% off all dairy products')
is_active	—	BIT	Flag: 1 if the promotion is currently active
days_until_start	—	INT	Days from today until start_date (negative if already started)
days_since_ended	—	INT	Days since the promotion ended (NULL if not yet ended)

PURCHASE_ORDER_LINES

Supply Chain

Line-item detail for every purchase order. Each row is one product line within a PO, capturing quantities, pricing and cost build-up.

Column	Role	Data Type	Description
po_line_key	PK	INT	Surrogate key for this PO line record
po_number	FK	VARCHAR	Foreign key to purchase_orders table

line_number	—	INT	Sequential position of this line within the PO
product_id	FK	VARCHAR	Foreign key to products table
quantity_ordered	—	INT	Number of units ordered on this line
unit_price_kes	—	DECIMAL	Agreed unit price per item in KES from the supplier
line_total_kes	—	DECIMAL	Total line cost: quantity_ordered × unit_price_kes
calculated_unit_price	—	DECIMAL	QA cross-check: line_total_kes / quantity_ordered
landed_cost_kes	—	DECIMAL	Total cost including freight, duty and other landed cost components
estimated_vat_kes	—	DECIMAL	Estimated VAT amount in KES based on applicable rate
po_prefix	—	VARCHAR	Prefix extracted from po_number (e.g. 'PO') for pattern analysis
po_sequence_number	—	INT	Numeric sequence from po_number for ordering
quality_flag	—	VARCHAR	Silver QA flag: PASS, WARN, or FAIL

PURCHASE_ORDERS

Supply Chain

Header record for each purchase order raised with a supplier. Tracks order lifecycle, expected delivery, lead time and action priority for procurement management.

Column	Role	Data Type	Description
po_number	PK	VARCHAR	Unique purchase order number — business-facing identifier
purchase_order_key	—	INT	Surrogate key for the PO header record
supplier_id	FK	VARCHAR	Foreign key to suppliers table
status	—	VARCHAR	Order status: Draft, Submitted, Confirmed, Partially Received, Received, Cancelled
status_category	—	VARCHAR	Simplified status grouping: Open, In-Progress, Closed
order_date	—	DATE	Date the purchase order was raised
expected_delivery_date	—	DATE	Anticipated delivery date agreed with the supplier
total_amount_kes	—	DECIMAL	Total value of all lines on the PO in KES
order_year	—	INT	Extracted year of order_date
order_month	—	INT	Extracted month number
order_quarter	—	VARCHAR	Quarter label (Q1–Q4)
order_year_month	—	VARCHAR	YYYY-MM string for period aggregation

expected_lead_time_days	—	INT	Number of days between order_date and expected_delivery_date
days_since_order	—	INT	Number of days elapsed since order_date as of silver load
po_prefix	—	VARCHAR	Prefix extracted from po_number
po_sequence_number	—	INT	Numeric sequence extracted from po_number
action_priority	—	VARCHAR	Derived flag for procurement team: Urgent, Monitor, On Track, Overdue
quality_flag	—	VARCHAR	Silver QA flag: PASS, WARN, or FAIL

RETURNS

Sales

Records of product returns linked to original POS transactions. Supports return rate analysis, refund tracking and product quality investigation.

Column	Role	Data Type	Description
return_id	PK	VARCHAR	Unique return event identifier
original_transaction_id	FK	VARCHAR	Foreign key back to pos_transactions — the original sale being returned
return_date	—	DATE	Date the return was processed
product_id	FK	VARCHAR	Foreign key to products — the returned product
quantity_returned	—	INT	Number of units returned
refund_amount	—	DECIMAL	Total refund value issued to the customer in KES
return_reason	—	VARCHAR	Reason given for the return: Defective, Wrong Item, Changed Mind, Damaged in Transit, etc.

SERVICE_INTERACTIONS

Customer Experience

Log of every customer service interaction across all channels. Enables service quality measurement, resolution time tracking and issue type analysis.

Column	Role	Data Type	Description
interaction_id	PK	VARCHAR	Unique identifier for this customer service interaction
customer_id	FK	VARCHAR	Foreign key to crm table
interaction_date	—	DATE	Date the interaction took place
channel	—	VARCHAR	Channel used: Phone, Email, Live Chat, In-Person, Social Media, WhatsApp
issue_type	—	VARCHAR	Category of the issue raised: Delivery, Billing, Product, Returns, General Inquiry
resolution_time_minutes	—	INT	Time taken to resolve the issue from first contact to closure, in minutes
satisfaction_score	—	INT	Post-interaction satisfaction rating given by customer (1–5 scale)

STORE_DAILY_FINANCIALS

Finance

Daily financial summary per store — sales, costs, margins and net profit. Provides a fast aggregate view without drilling into individual GL transactions.

Column	Role	Data Type	Description
store_id	FK	VARCHAR	Foreign key to stores table
date	—	DATE	The calendar date for this financial summary
sales_kes	—	DECIMAL	Total gross sales revenue for the store on this date in KES
cost_of_goods_sold	—	DECIMAL	COGS for the day — direct cost of products sold
gross_margin	—	DECIMAL	Gross profit: sales_kes minus cost_of_goods_sold
operating_expenses	—	DECIMAL	Operating costs for the day (rent, utilities, labour, etc.)
net_profit	—	DECIMAL	Bottom-line profit: gross_margin minus operating_expenses

STORES

Operations

Master reference for all retail store locations including their lifecycle, format, size and status. Supports SCD Type 2 via valid_from/valid_to for stores that have changed attributes over time.

Column	Role	Data Type	Description
store_id	PK	VARCHAR	Unique store identifier — joins to all sales, inventory, HR and finance tables
valid_from	—	DATE	SCD effective start date for this version of store attributes
valid_to	—	DATE	SCD effective end date; NULL if this is the current active record
store_name	—	VARCHAR	Display name of the store (e.g. 'Westgate Nairobi')
county	FK	VARCHAR	Kenyan county the store is located in — links to gis_counties
town	—	VARCHAR	Town or suburb within the county
format	—	VARCHAR	Store format: Superstore, Express, Convenience, Online Fulfilment Centre
size_sqm	—	DECIMAL	Floor area of the store in square metres
opening_date	—	DATE	Date the store first opened to the public
closing_date	—	DATE	Date the store closed permanently; NULL if currently open
is_active	—	BIT	Flag: 1 if the store is currently operational

SUPPLIERS

Supply Chain

Supplier master reference including contact details, payment terms and product category specialisation. Supports SCD Type 2 for suppliers whose attributes change.

Column	Role	Data Type	Description
supplier_id	PK	VARCHAR	Unique supplier identifier
valid_from	—	DATE	SCD effective start date for this supplier version
valid_to	—	DATE	SCD effective end date; NULL if current record
supplier_name	—	VARCHAR	Trading name of the supplier company
contact_person	—	VARCHAR	Primary contact person's full name at the supplier
phone	—	VARCHAR	Supplier contact telephone number
email	—	VARCHAR	Supplier contact email address
payment_terms	—	VARCHAR	Agreed payment terms: Net 30, Net 60, COD, Prepaid, etc.
lead_time_days	—	INT	Standard number of days from order to delivery for this supplier
category	—	VARCHAR	Product category the supplier primarily provides
tax_id	—	VARCHAR	Supplier KRA PIN or tax registration number

TIME_TRACKING

Human Resources

Raw clock-in/clock-out event log for employees. Each row is a single timestamped event (Check In or Check Out). Used for attendance analysis, shift compliance and overtime calculation.

Column	Role	Data Type	Description
event_id	PK	INT	Surrogate key for the time-tracking event
employee_id	FK	VARCHAR	Foreign key to hr table
store_id	FK	VARCHAR	Foreign key to stores — where the clock event occurred
timestamp	—	DATETIME	Full datetime of the clock event
date_	—	DATE	Date component extracted from timestamp
time_	—	TIME	Time component extracted from timestamp
event_type	—	VARCHAR	Type of event: Check In or Check Out